AI 300 Logistic Regression, Assignment *

Beaver-Edge AI Institute

1. Consider the following **Sigmoid function**:

$$\sigma(z) = \frac{1}{1+\exp(-z)}, \ \forall\ z \in \mathbb{R}.$$

   (a) What is the range of $\sigma(z)$?
   (b) Compute $\sigma(-\infty)$, $\sigma(\infty)$, $\sigma(0)$.
   (c) Compute $1-\sigma(z)$.
   (d) Compute $\frac{d\sigma(z)}{dz}$. Write your solution in terms of $\sigma(z)$.
   (e) Compute $\frac{d^2\sigma(z)}{dz^2}$. Write your solution in terms of $\sigma(z)$.
   (f) Discuss the monotonicity property of $\sigma(z)$.
   (g) Plot $\sigma(z)$.

2. Consider the following binary classification model:

$$\hat{y} = \begin{cases} 1 & \text{with probability } \sigma\left(\theta^\top \bar{\mathbf{x}}\right) \\ -1 & \text{with probability } 1-\sigma\left(\theta^\top \bar{\mathbf{x}}\right) \end{cases}.$$

   The ground-truth output values are 1 and -1.

   (a) Write down the cross entropy loss function, denoted as L_θ.
   (b) Compute $\nabla_\theta\ L_\theta$.
   (c) Compute $\nabla_\theta^2\ L_\theta$.
   (d) Prove that $\nabla_\theta^2\ L_\theta$ is positive semi-definite.

3. Consider the following binary classification model:

$$\hat{y} = \begin{cases} 1 & \text{with probability } \sigma\left(\theta^\top \bar{\mathbf{x}}\right) \\ 0 & \text{with probability } 1-\sigma\left(\theta^\top \bar{\mathbf{x}}\right) \end{cases}.$$

   The ground-truth output values are 1 and 0.

   Define a class called `MyBinaryClassLogisticRegression`.

   (a) The attribute is θ.
   (b) Methods include `fit`, `predict`, `score`. All these mimic those Sklearn models, except `score` has an input that allows you to choose between outputting an average score or an f1 score.

4. (a) Find any dataset for binary classification (e.g., Breast Cancer Wisconsin, Titanic, Pima Indians Diabetes).
   (b) Pre-process data in an appropriate way (e.g., removing missing data, one-hot encoding for categorical data, normalizing numerical features).
   (c) Split the dataset into the training dataset and test dataset.
   (d) Train your model defined in Problem 3.
   (e) Test your model with the test dataset.

5. There exists a function $f : \mathbb{R}^K \to \mathbb{R}$, such that

$$\nabla_{\mathbf{z}} \ f(\mathbf{z}) = \text{Softmax} \ (\mathbf{z}).$$

   What is f?

6. Mathematically compute

$$\frac{\partial \ \text{Softmax}_i(\mathbf{z})}{\partial z_j}.$$

   Your solution should be expressed in terms of the softmax function.

7. This is a coding task.

   Let N be the number of samples and K be the number of labels.

   Let $\mathbf{z}$ be with shape `(N,K)`.

   (a) Write code to compute $\text{Softmax}(\mathbf{z})$, denoted as `softmax_val`. The shape is `(N,K)`.
   (b) Write code to compute $\nabla_{\mathbf{z}} \ \text{Softmax}(\mathbf{z})$, denoted as `D_softmax_val`. The shape is `(N,K,K)`. Thus, `D_softmax_val[n,i,j]` refers to $\frac{\partial \ \text{Softmax}_i\left(\mathbf{z}^{(n)}\right)}{\partial z_j}$.

   In your solution, avoid using any loop.

8. Consider the following multi-class logistic regression model.

   Let $\bar{\mathbf{x}} \in \mathbb{R}^d$. Let $\theta_{\mathbf{k}} \in \mathbb{R}^d$ for $k \in \{0, \cdots, K-1\}$.

   Denote

$$\mathbf{\Theta} = \begin{bmatrix} \theta_{\mathbf{0}}^\top \\ \theta_{\mathbf{1}}^\top \\ \vdots \\ \theta_{\mathbf{K-1}}^\top \end{bmatrix} \in \mathbb{R}^{K \times d}.$$

Then the model predicts

$$\hat{y} = k \text{ with probability } \mathrm{Softmax}_k \left(\mathbf{\Theta} \bar{\mathbf{x}} \right).$$

(a) Write down the cross-entropy loss function, denoted as $L_{\mathbf{\Theta}} = \frac{1}{N} \sum_{n=0}^{N-1} L_{\mathbf{\Theta}}^{(n)}$.

(b) Mathematically compute $\nabla_{\mathbf{\Theta}} L_{\mathbf{\Theta}}$. The shape is `(K,K)`.

9. In this problem, you are asked to prove that $L_{\mathbf{\Theta}}^{(n)}$ is positive semi-definite.

   Hint: If $f(\mathbf{x})$ is a convex function, then $f(\mathbf{A}\mathbf{x} + \mathbf{b})$ is also convex.

10. Define a class called `MyMultiClassLogisticRidgeRegression`.

    (a) In training, the loss function is the sum of the cross-entropy and ridge regularization (not regularized on intercepts).

    (b) The attributes are $\mathbf{\Theta}$ (learnable parameters) and weight decay (hyperparameter).

    (c) Methods include `fit`, `predict`, `score`. All these mimic those Sklearn models, except `score` has an input that allows you to choose between outputting an average score or an f1 score.

11. (a) Find any dataset for multi-class classification (e.g., MNIST, CIFAR10).

    (b) Pre-process data in an appropriate way (e.g., one-hot encoding for categorical data, normalizing numerical features).

    (c) Split the dataset into the training dataset and test dataset.

    (d) Train with the multi-class logistic regression with ridge regularization model, defined in Problem 10.

    (e) Do K-fold cross validation on the above training dataset to optimize over the hyperparameter (weight decay).

    (f) After optimizing your hyperparameter and learnable parameters, compute the accuracy score and the f1 score with the test dataset.

12. Define

$$f_i = \frac{\exp\left(z_i / \tau\right)}{\sum_{j=0}^{N-1} \exp\left(z_j / \tau\right)}.$$

(a) Compute

$$\lim_{\tau \to 0^+} f_i.$$

(b) Compute

$$\lim_{\tau \to \infty} f_i.$$

(c) Explain the intuition of your results above.